Math 252
Spring 2021

Name ______________________ Section: A B

Test 2

Instructions

1. As you do the problems on this test, be sure to show your work.
2. If you use your calculator, enter both what you typed into your calculator and the result.
3. If you round, round late and show the value (3 or 4 significant digits) both before and after rounding.

1 Suppose $p(A) = 0.5$, $p(B) = 0.3$, and $p(A \cap B) = 0.1$.

a) What is $p(A \cup B)$?

b) What is $p(A \mid B)$?

c) What is $p(B \mid A)$?

d) Are events A and B **independent**? Explain how you know.

e) Are events A and B **mutually exclusive**? Explain how you know.

2 Sally is taking a multiple-choice quiz that has 10 questions on it. Each question has 4 possible answers, of which only one is correct. Suppose Sally just guesses randomly on each question.

a) What is the probability that Sally gets **exactly 3** correct?

b) What is the expected value for the number she will get correct?

3 Each of these questions deals with probabilities when **four 10-sided dice** are rolled. Express each answer in decimal notation.

a) How many dice are being rolled?

b) How many sides does each die have?

c) What is the probability that the four numbers rolled will **all be the same**?

d) What is the probability that the four numbers rolled will be **distinct**? (four different numbers, no repetitions)

e) What is the probability that at least one 1 and at least one 2 are rolled?

4 One of the variations of the the Michigan Lottery Daily 4 game has two possible prizes. The big prize is \$3200. The probability of winning this prize is 1/10000. The smaller prize is \$360. The probability of winning this prize is 1/2500.[1] Of course, if you don't win one of those prizes, you get nothing.

Let W be the amount of money won from one of these lottery tickets.

a) What is $p(W = 0)$?

b) What is the expected value of W?

c) What does the expected value of W tell us about this lottery game. (Note: It costs \$1 to buy one of these tickets).

[1] This information is available on their website in a different form – and has some errors because they mix up odds and probability. I've fixed those errors here.

5 The Sieben Siegel (Seven Seals) card game uses a non-standard deck. The deck has **nine numbers** (1 – 9) in each of **five colors** (red, green, blue, yellow, and purple).

For this problem, you do not need to compute a numerical answer if you don't want to. Just write down the expression that gives the answer. You are, of course, welcome to compute the answer numerically as well.

a) How many cards are in the deck?

b) If you are dealt six cards, what is the probability that they are **all the same color**?

c) If you are dealt six cards, what is the probability that you have **3 pairs**? (2 of one number, 2 of a different number, and 2 of a third number)

d) If you are dealt six cards, what is the probability that you have **3-of-a-kind**? (3-of-a-kind means that 3 cards are the same number, and the other three are different from each other and from the three that match.)

Go back and make sure you didn't slip back into standard deck mode anywhere.

6 Calvin has three hats. Each hat contains some maroon and some gold marbles.

- Hat A has 1 maroon and 4 gold.
- Hat B has 1 maroon and 1 gold.
- Hat C has 9 maroon and 1 gold.

Calvin randomly selects a hat (with equal probability) and then randomly selects a marble from that hat.

a) Alltogether, there are 17 marbles. Explain why they are not equally likely to be selected.

b) What is the probability that the selected marble comes from hat A?

c) What is the probability that the selected marble is the maroon marble in hat A?

d) What is the probability that the selected marble is maroon (from any hat)?

e) If the selected marble is maroon, what is the probability that the hat selected is hat A?

f) Let M be the event that the marble selected is marroon. Let A be the event that the hat selected is hat A. Circle the correct notation for the probabilities in parts b – e above.

part b: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

part c: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

part d: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

part e: $p(A)$ $p(M)$ $p(A \cap M)$ $p(A \cup M)$ $p(A \mid M)$ $p(M \mid A)$

Space for additional work.